of food borne diseases. Microbiologists are currently employed in all food and dairy processing industries. They are also employed in Mineral water companies to check the quality of water.

**Agricultural Microbiology** – It is concerned with the impact of microorganisms on agriculture. Most bacteria and fungi live saprophytically on dead and organic matter of the soil. They decompose the complex organic matter into simpler form making it available for the soil microorganisms. Thus they form an important constituent of soil called humus. Certain microbes increase the fertility of the soil by converting the atmospheric nitrogen into ammonia, nitrites and nitrates. This is brought about by the microbes such as *Nitrosomonas*, *Nitrobacter* and *Rhizobium* sp. Agricultural microbiologists try to combat plant diseases that attack commercial food crops and they also work on methods to increase soil fertility and crop yields.

**Industrial Microbiology** – Microorganisms are used to make products such as antibiotics, vaccines, steroids, alcohols, vitamins, amino acids and enzymes. Some important drugs are synthesized by microorganisms such as streptomycin, penicillin, chloramphenicol, tetracycline. Industrial microbiologists work on improving the strains that produce the industrially important products and thereby increase the yield. The Research and Development (R&D) units in the industries provide various job opportunities to microbiologists.

**Directors of Research Units and Universities** – Many microbiologists work for universities, where they teach and do research. Microbiologists can become directors of research in medical centres, private firms, or government agencies. Those who hold a teaching and research position in a university can advance to the rank of full professor. They can also make significant discoveries in their research and gain the recognition of other microbiologists. Many scientists consider this to be the highest form of advancement.

**Microbial Genetics and Molecular Biology** – The use of micro organisms has been very helpful in understanding the functions of the genes. Microbial geneticists play an important role in applied microbiology by producing new microbial strains that are more efficient in synthesizing useful products. Genetic techniques are used to test substances for their ability to cause cancer. Microbiologists are in greater demand in genetic engineering companies and research units.

**Biomining** – Microbes are used in extracting valuable metals like uranium from rocks. *Thiobacillus ferrooxidans* unlocks energy from inorganic compounds like iron sulphide. During this process, it produces sulphuric acid and iron sulphate. The use of micro organisms in mining has considerably reduced the cost of mining to 75%. Microbiologists involved in Biomining research are highly paid in the Government sector.

**Medical coding** – Medical coding is the transformation of healthcare diagnosis, procedures, medical services and equipment into universal medical alphanumeric codes. The diagnoses and procedure codes are taken from



TN_GOVT_XI_Micro_Biology_FM.indd 6 16-02-2019 10.43.10 AM